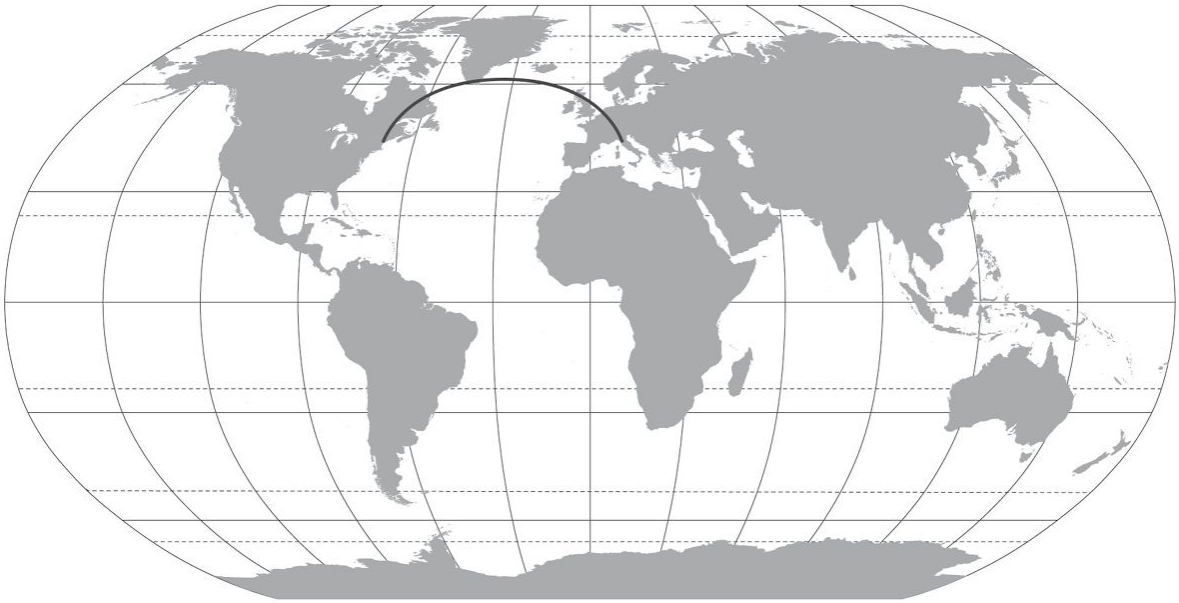


Figure 3.9 The further north you go, the smaller the distance between two meridians.

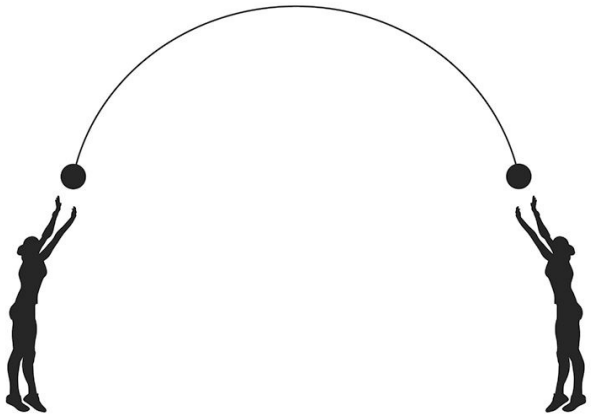


Figure 3.10 The higher up something is, the more quickly time passes for it.

When I was a university student, black holes were regarded as a scarcely credible implication of an esoteric theory. Today they are observed in their hundreds and studied in detail by astronomers. One of these black holes, with a mass a million times greater than the Sun, is located at the centre of our galaxy – we can observe stars orbiting around it. Some, passing too close, are destroyed by its violent gravity.

Further still, the theory predicts that space ripples like the surface of the sea, and that these ripples are waves similar to the electromagnetic ones which make television possible. The effects of these ‘gravitational waves’ can be observed in the sky on binary stars: they radiate such waves, losing energy and slowly falling towards each other.[fn13] Gravitational waves produced by two black holes falling into one another were directly observed by an antenna